Can Meaning Have Universal Coordinates? A Negative-and-Positive Result

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Abstract

We tried to build a *universal coordinate system for meaning*: a frozen, open, hierarchical code (e.g. `JYZA-7M-X0-A8`) that any text in any language would receive, reachable from many embedding models through linear adapters, so that translations, paraphrases and different models agree on the same address. Before running the decisive experiments we fixed a four-part pass/fail gate on held-out data (coverage, translation agreement, cross-model agreement, language collapse), and later amended it to close a loophole: the gate must be met with at least 256 top-level cells. **No configuration we built passes.** The frozen release (4096 top-level cells, multilingual-e5-small anchor) fails all four criteria (coverage 29%, confident translation agreement 65%, cross-model 16%, worst language collapse 34%). Agreement is bought with coarseness: each 4× finer top level costs roughly 10–25 points on the gate metrics, and every near-pass uses 32–64 cells. The anchor matters (LaBSE beats e5-small by 11 points of coverage and 4.5 points of confident translation agreement at the same granularity) but not enough to change the conclusion: a GPU run at ~450k sentences with LaBSE and bge-m3 anchors removes language collapse and reaches 88% confident translation agreement at 256 cells, yet cross-model agreement peaks at 50% (bar: 60%). We also document a *language-collapse* failure: at ≥ 256 cells, k-means forms cells that are mostly one language (up to 55% of Portuguese test sentences), which a diversified sample only partly fixes. What does work: (1) *coarse regions* (32–64 cells) where translations agree 78–89% of the time when confident, useful for multilingual topic analytics, routing and caching; and (2) *migration through stored canonical vectors*: an index built with one model can be queried with another at 96.5–99.1% recall@10 without re-embedding. We release everything, including the gate, the negative results and the scripts to rerun them.

1. Introduction

Embedding vectors are a good *computational* representation of meaning and a poor *data format*: a vector from model A means nothing to model B, it cannot be read or grouped in SQL, and every model upgrade forces re-embedding the corpus. A natural ambition is a shared, stable *address* for meaning, like latitude and longitude: readable, hierarchical, frozen, and the same whichever language or model produced the text.

This paper asks whether that ambition is achievable with current multilingual encoders and simple, transparent machinery (linear projections, residual k-means, linear adapters). Our answer is split:

- **Negative.** A *fine-grained* universal code (hundreds to thousands of top-level cells, stable across languages and models) is not supported by our measurements. The failure is structural, not a tuning issue: any partition of a continuous space splits equivalent texts near its boundaries, and the fraction split grows quickly with the number of cells.
- **Positive.** A *coarse* shared partition is stable enough to be useful, and the continuous canonical space behind it is an effective hub for cross-model migration.

We think the negative result is worth publishing on its own. “Semantic IDs” and shared embedding spaces are active topics, and claims of a universal meaning code are easy to make on in-sample numbers; ours looked good in-sample too (81% of training sentences get a confident top-level code versus 26% held out).

The project name comes from Bhartṛhari’s *sphoṭa* (5th c. CE): meaning that bursts forth whole and stays the same across its utterances. Our results suggest it stays the same only at a coarse resolution.

Contributions. 1. A complete, open method for a frozen hierarchical meaning code (§3). 2. A pre-registered decision gate, plus an amendment closing a granularity loophole (§4). 3. Full gate results across granularity, sample design and anchor (§5). 4. The granularity trade-off, the anchor effect and the language-collapse failure (§5–6). 5. Evidence for what does work: coarse regions and vector-based migration (§7).

2. Related work

Semantic IDs. TIGER (Rajput et al., 2023) quantizes item embeddings with an RQ-VAE into codeword tuples for generative retrieval; DSI (Tay et al., 2022) uses hierarchical k-means document IDs; Singh et al. (2024) study Semantic IDs for ranking. These codes are private to one system and refit per catalogue, so they never face the cross-language and cross-model stability test that a *universal* code must pass. Codebook collapse in RQ-VAE is mitigated by EMA updates and dead-code revival (Zeghidour et al., 2021; Lee et al., 2022).

Embedding translation. vec2vec (Jha et al., 2025) translates between embedding spaces without paired data, in support of the Platonic representation hypothesis (Huh et al., 2024). Our results are compatible with that hypothesis at the level of *continuous* geometry (migration works) while showing that *discrete* agreement is far weaker.

Language-agnostic spaces. LASER, LaBSE (Feng et al., 2022) and SONAR (Duquenne et al., 2023) embed many languages jointly; Large Concept Models (2024) operate in SONAR space. We quantify how much of their cross-lingual alignment survives discretization.

Topic models. BERTopic (Grootendorst, 2022) clusters embeddings and labels clusters with c-TF-IDF, refitting per corpus. Our positive result is close to a *frozen, multilingual, shared* BERTopic-style partition.

3. Method

3.1 Anchor encoder

`intfloat/multilingual-e5-small` (MIT, 384-d, ~100 languages), input `"query: " + text`, mean pooling, L2 normalization. It was chosen in a bake-off against bge-m3, LaBSE and paraphrase-multilingual-MiniLM for speed, size and an exact ONNX export (max abs diff $7 \cdot 10^{-8}$ versus PyTorch), accepting slightly weaker translation alignment than LaBSE (0.67 vs 0.62 L1 agreement at 256 cells in the bake-off). §5.3 revisits this choice.

3.2 Canonical space (meaning-LDA)

With unit vectors `x`, `y = (x − μ) W`: PCA to 384 directions, then the generalized eigenvectors of `(S_total, S_within + reg)`, where `S_within` is the scatter of translations of the same sentence (FLORES-200 dev, 800 meanings × 12 languages; 7.2k Tatoeba pairs), shrunk toward isotropy (`reg = 0.05`). The top 256 directions are scaled so within-meaning noise has unit variance. As an ablation, PCA alone gives 34.1% cross-lingual L1 agreement versus 46.1% with meaning-LDA (4096 cells).

3.3 Quantizer and code

Residual k-means with 4 levels (4096·256·256·256) on 77.6k sentences (Wikipedia sample in 12 languages + FLORES dev). Codes are Crockford base32 (widths 4/2/2/2) after a published affine permutation, so 85% of single-symbol typos are rejected without a checksum.

3.4 Confidence

At each level, a softmax over the 16 nearest centroids' squared distances, with a temperature fitted by maximum likelihood to predict the *held-out translation's* symbol; the product over levels is the prefix confidence. Confidence drives truncation, alternates and the gate's "confident portion".

3.5 Adapters

Other models enter the canonical space through ridge maps fitted on 40k paired texts (registry: all-MiniLM-L6-v2, bge-small-en-v1.5, gte-small, paraphrase-multilingual-MiniLM-L12-v2, LaBSE). An unsupervised vec2vec-style adapter failed at our CPU budget (top-1 retrieval 0% versus 96.5% for ridge on the same texts), so supervised adapters are the default.

4. The decision gate

Owner review of the first release concluded that in-sample numbers did not support a “GPS for meaning” claim. We therefore fixed the following gate **before** any v2 run (D-017). All criteria use held-out FLORES-200 devtest sentences in 12 languages, never used to fit the space, codebooks, calibration, labels or adapters.

#	Criterion	Definition	Bar
1	Coverage	share of held-out sentences (all 12 languages) with L1 confidence ≥ 0.7	≥ 60%
2	Translation agreement	among en ↔ x pairs whose English code has L1 confidence ≥ 0.7, share with the same L1 code (mean over 11 languages)	≥ 80%
3	Cross-model agreement	same English text, certified adapter vs anchor, same L1 code (mean over adapters)	≥ 60%
4	Language collapse	per language, share of its sentences in L1 cells whose training members are ≥ 80% that language (English excluded)	≤ 5% for every language

Amendment (D-018). A tiny-scale dry run “passed” all four criteria with 16 top-level cells and 6 languages. A gate that can be passed by making the partition coarse says nothing about coordinates. We amended it: the gate must be met with **≥ 256 top-level cells**, on all 12 languages, with the cell count declared before the run. Results at 32–64 cells are reported as evidence for a *coarse topic* product only.

5. Results

5.1 Full gate results

Build	L1 cells	Coverage (≥ 60%)	Translation, confident (≥ 80%)	Cross-model (≥ 60%)	Worst collapse (≤ 5%)	Pass
v1 frozen release	4096	29.4%	65.1%	15.7%	33.8% (pt)	0/4
v1 re-clustered	1024	42.6%	69.7%	23.1%	44.2% (pt)	0/4
v1 re-clustered	256	49.2%	74.4%	32.8%	54.6% (pt)	0/4
v1 re-clustered	128	54.1%	73.6%	44.6%	0.1%	1/4
v1 re-clustered	64	65.0%	78.3%	57.1%	0.0%	2/4
v1 re-clustered	32	69.0%	79.3%	64.5%	0.0%	3/4
v2-mini (diversified sample)	4096	30.9%	65.0%	16.3%	26.8% (pt)	0/4
v2-mini	32	63.9%	78.8%	62.3%	0.1%	3/4
e5-small anchor (20k corpus)	32	55.8%	84.2%	60.3%*	0.0%	3/4
LaBSE anchor (20k corpus)	32	66.5%	88.7%	53.0%*	0.0%	3/4
LaBSE anchor (20k corpus)	64	64.4%	85.3%	39.1%*	0.0%	3/4

* Reduced cross-model panel (the other anchor and all-MiniLM-L6-v2). Sources: `bench/results/l1-sweep.json`, `v2-mini-gate.json`, `v2-mini-4096-gate.json`, `anchor-check.json`.

No configuration passes all four criteria at any granularity, and none passes more than one criterion at the amended bar of ≥ 256 cells: every build at 256, 1024 or 4096 cells fails all four. The closest configurations (3 of 4) all use 32–64 top-level cells, which by the amendment cannot support a coordinate-system claim.

5.2 The granularity trade-off

Stability is bought with coarseness. Along the v1 re-clustering sweep, each 4× increase in top-level cells costs roughly 10–25 points on the gate metrics:

Step	Coverage	Translation (confident)	Cross-model
32 → 128	−14.9	−5.7	−20.0
64 → 256	−15.8	−3.9	−24.3
256 → 1024	−6.6	−4.7	−9.8
1024 → 4096	−13.2	−4.6	−7.3

Cross-model agreement degrades fastest, then coverage; confident translation agreement degrades more slowly because the confidence filter removes the hardest boundary cases (at the price of coverage). The same pattern holds for LaBSE (32 → 64 cells: −2.1 coverage, −3.4 translation, −13.9 cross-model). We see no sign of a plateau: finer partitions simply place more boundaries through regions where equivalent texts land.

5.3 The anchor effect: LaBSE vs e5-small

On the same 20k corpus at 32 cells, LaBSE beats e5-small on coverage (66.5% vs 55.8%, +10.7) and confident translation agreement (88.7% vs 84.2%, +4.5), and is the only build that clears the translation bar. It is worse on cross-model agreement with the reduced panel (53.0% vs 60.3%), plausibly because the panel's other models are closer to e5's geometry. The anchor is therefore a first-order factor, comparable in size to one 4× granularity step, but it does not change the conclusion: a better anchor buys roughly one step of resolution, not the six steps from 32 to 4096 cells. We tested bge-m3 and LaBSE as anchors on a GPU in §5.5.

5.4 Language collapse

The first release gave confident top-level codes to only 4.5% of Portuguese held-out sentences versus 21–36% for other languages. Data, language tags, alignment (en→pt nearest neighbour 99.2%) and encoding were correct. The cause: **52.8% of Portuguese test sentences fell into Portuguese-dominated cells** (80–96% Portuguese training members; 5 cells hold 53% of pt test sentences), regardless of topic; for every other language the share was ≤ 0.2% (`bench/results/language-collapse-small.json`). The Portuguese Wikipedia sample is unusually homogeneous (mean pairwise cosine 0.84 vs ~0.78, dominated by sports and list stubs), so k-means spent cells on *language identity* instead of meaning.

Two findings generalize:

- **Collapse is a function of granularity.** It is absent at ≤ 128 cells (≤ 0.1%) and appears abruptly at 256 cells (54.6% for pt). Once the partition is fine enough, residual language signal in the anchor becomes a cheaper way for k-means to reduce distortion than topic.
- **The sample explains only part of it.** A diversified per-language sample (≤ 40 sentences per title category, near-duplicates removed, ≤ 8000 per language) cut the worst collapse at 4096 cells only from 33.8% to 26.8%. Part of the cause is the anchor itself: e5-small keeps a separable Portuguese direction that meaning-LDA does not fully remove.

Language collapse is invisible in average cross-lingual agreement and should be reported as a separate metric by any multilingual quantization work.

5.5 GPU run: stronger anchors at medium scale

One bounded run (Kaggle T4, 12 languages, ~450k sentences, diversified per-language sample), evaluated on held-out FLORES-200 devtest against the pre-registered bar:

anchor	L1 regions	coverage ≥ 60%	translation (confident) ≥ 80%	cross-model ≥ 60%	collapse ≤ 5%	passes
multilingual-e5-small	256	46.3% FAIL	74.4% FAIL	31.3% FAIL	40.9% FAIL	0/4
multilingual-e5-small	1024	39.3% FAIL	71.6% FAIL	24.7% FAIL	31.6% FAIL	0/4
multilingual-e5-small	4096	26.7% FAIL	66.0% FAIL	16.2% FAIL	34.2% FAIL	0/4
LaBSE	256	69.4% PASS	83.5% PASS	40.1% FAIL	0.2% PASS	3/4
LaBSE	1024	55.2% FAIL	83.9% PASS	29.3% FAIL	0.1% PASS	2/4
LaBSE	4096	46.9% FAIL	78.4% FAIL	17.7% FAIL	0.4% PASS	1/4
bge-m3	**256**	67.2% PASS	88.0% PASS	**50.2% FAIL**	0.1% PASS	3/4
bge-m3	1024	62.8% PASS	82.6% PASS	38.0% FAIL	0.1% PASS	3/4
bge-m3	4096	54.5% FAIL	78.6% FAIL	25.3% FAIL	0.2% PASS	1/4

The stronger anchors remove language collapse (≤ 0.4% vs 34–41% for e5-small) and lift confident translation agreement to 88.0% (bge-m3, 256 cells). Cross-model agreement does not follow: 50.2% at best, falling 50 → 38 → 25% with each 4× refinement. **The gate fails**, as we predicted. The bottleneck is not the anchor or the data volume but the linear adapters between models.

6. Why fine-grained universal codes fail

Three effects compound. (1) **Boundary mass:** equivalent texts are not identical points but small clouds (translation noise is our unit of distance); the fraction of clouds cut by some cell boundary rises with the number of cells. (2) **Adapter error:** a linear adapter adds its own displacement, which at fine granularity exceeds the cell radius; hence cross-model agreement falls fastest. (3) **Residual language signal:** at fine granularity, k-means can profitably spend cells on language rather than meaning. Confidence and alternates make (1) visible but do not remove it; they trade coverage for precision. A code can be calibrated and honest about its uncertainty and still not be *universal*.

7. What works

7.1 Coarse regions

At 32–64 top-level cells, confident translations share a region 78–89% of the time and there is no language collapse; with the v1 space or the LaBSE anchor, coverage is 64–69%, and cross-model agreement is 57–65% for the v1 space (lower for LaBSE at 64 cells: 39%, reduced panel). That is a useful frozen, multilingual, shared topic partition: readable region codes for analytics dashboards, SQL `GROUP BY`, routing queries to specialised handlers, and semantic caching keys, with finer levels kept as optional detail carrying explicit confidence.

7.2 Migration through stored canonical vectors

The continuous canonical space is a much better hub than the discrete code. Documents indexed with model X through its adapter and queried with the anchor reach recall@10 of 96.5% (all-MiniLM-L6-v2), 99.1% (bge-small-en-v1.5), 99.0% (gte-small) and 98.5% (paraphrase-multilingual-MiniLM), versus 100% for re-embedding, and 98.3% for LaBSE in the paraphrase setting; using codes alone gives only 10–28%. Storing the 256-d

canonical vector next to each item lets an organisation change embedding models without re-embedding its corpus, with N adapters instead of N^2 pairwise maps.

7.3 Other measured properties of the frozen release

Cross-lingual L1 agreement 46.1% (stable mode 46.7%), paraphrase L1 agreement 71.6%; when both codes have L1 confidence ≥ 0.7 (16% of pairs), translations agree 88.7%, at ≥ 0.9 (5%), 96.5%. 92% of L1 cells are used on the training corpus (perplexity 2182 of 4096). Codes alone are a coarse index: translation retrieval top-1 30% (raw vectors 99.6%), SIB-200 topic P@10 16.6% vs 19.5%. Adding cross-model scatter to `S_within` helped at 1024 cells but not at 4096, while lowering cross-lingual agreement; we kept it off.

8. Limitations

Most experiments are CPU, `small` scale (≤ 80k fitting sentences, 12 languages, FLORES-200 devtest for evaluation); one GPU run reached `medium` scale (~450k sentences, §5.5) and confirmed the failure at ≥ 256 cells. We did not try nonlinear or jointly trained adapters. The cross-model panel for the anchor comparison is reduced. Labels for cells are machine-generated and filtered, not natively reviewed. FLORES sentences are short, formal and topically narrow; real traffic is likely harder.

9. Conclusion

Can meaning have universal coordinates? At fine resolution, with today's small multilingual encoders and linear machinery, our pre-registered answer is **no**: no build passes a fixed, held-out gate at ≥ 256 cells, every 4× refinement costs 10–25 points, and fine partitions start encoding language instead of meaning. At coarse resolution the answer is a qualified **yes**: a few dozen shared regions are stable across languages and reasonably stable across models. And the continuous space behind the code is a practical hub for moving between models. We recommend that work on semantic IDs and shared meaning spaces report held-out, cross-language and cross-model agreement *as a function of granularity*, together with a language-collapse metric, and fix the bar before running.

Reproducibility

Gate definitions: `DECISIONS.md` D-017/D-018. Results: `bench/results/`. Experimental v2 build: `experiments/v2-mini/`. GPU run (§5.5): `notebooks/kaggle_gpu_gate.ipynb`, or `bash scripts/gpu_build_v2.sh --scale medium --sizes "256 1024 4096"`; results `bench/results/v2-gate-medium-*.json`. Code, artifacts and data recipes are Apache-2.0.

References

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